

# BIODROUGHT: a macroinvertebrate-based index to identify antecedent drying

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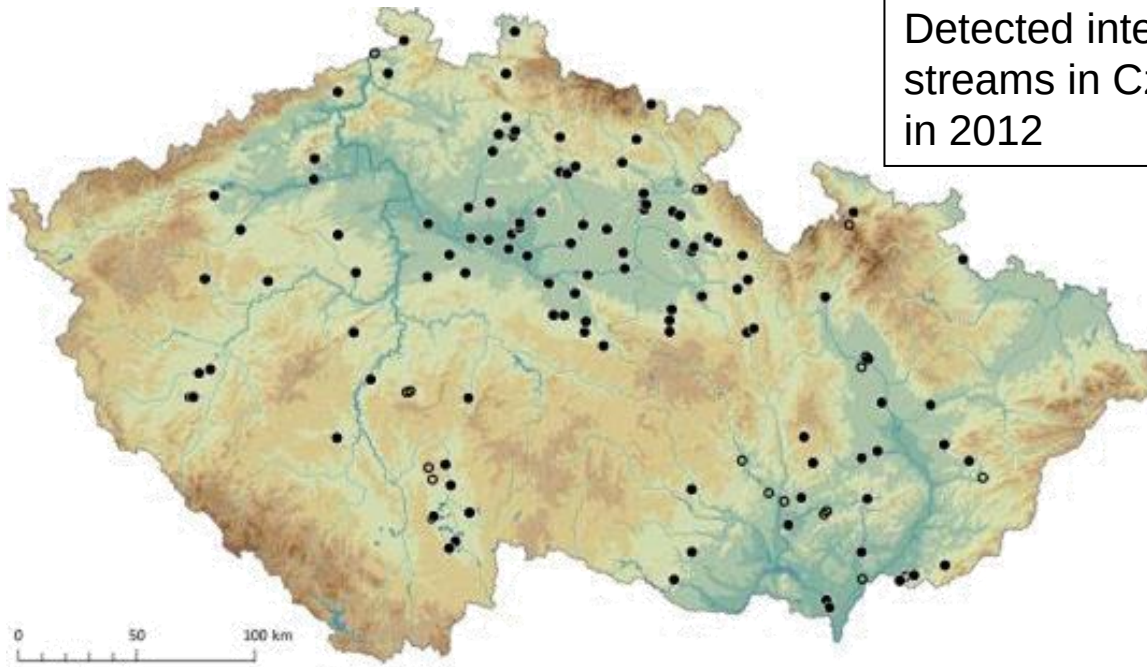
<sup>4</sup> University of Natural Resources and Life Sciences, Vienna, Austria

<sup>5</sup> University of Münster, Zoological Institute, Münster, Germany

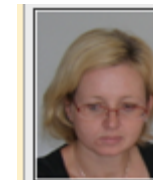
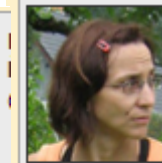
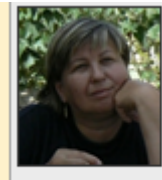


# BIODROUGHT project

- BIODROUGHT project run in 2012-2015, leader Dr. Pařil
- research team was established and a lot of theoretical and empirical knowledge about IRES gained
- a new tool to identify previous stream drying with use of benthic invertebrates was developed – BIODROUGHT index
- during SMIRES (2016-2020) the BIODROUGHT index was validated and published during cooperation with R. Stubbington and T. Datry

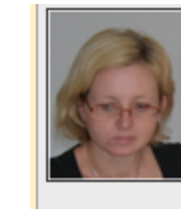
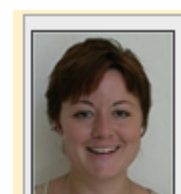
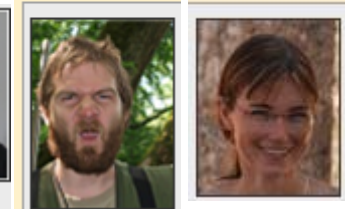
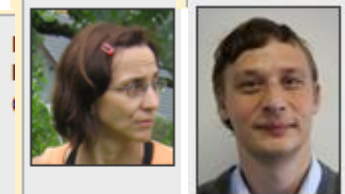
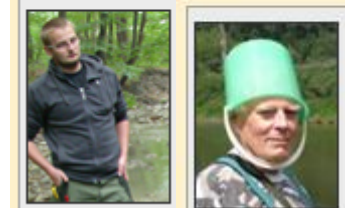
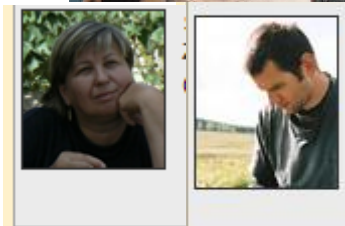


Detected intermittent streams in Czech Republic in 2012



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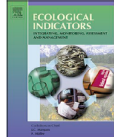


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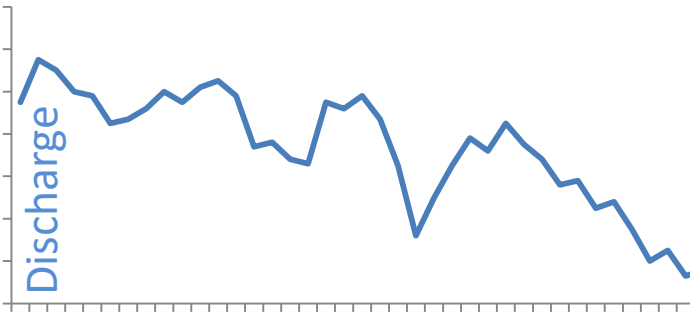
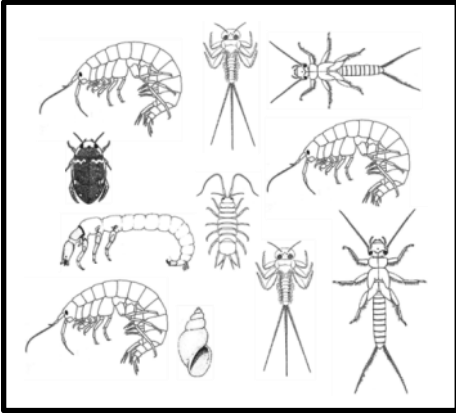
Recognition of stream drying based on benthic macroinvertebrates: A new tool in Central Europe

Michal Straka<sup>a,b,\*</sup>, Marek Polášek<sup>a,b</sup>, Vít Syrovátka<sup>b</sup>, Rachel Stubbington<sup>c</sup>, Světlana Zahradková<sup>b</sup>, Denisa Němejcová<sup>a</sup>, Lenka Šikulová<sup>b</sup>, Pavla Řezníčková<sup>b</sup>, Libuše Opatřilová<sup>a,d</sup>, Thibault Datry<sup>e</sup>, Petr Pařil<sup>b</sup>

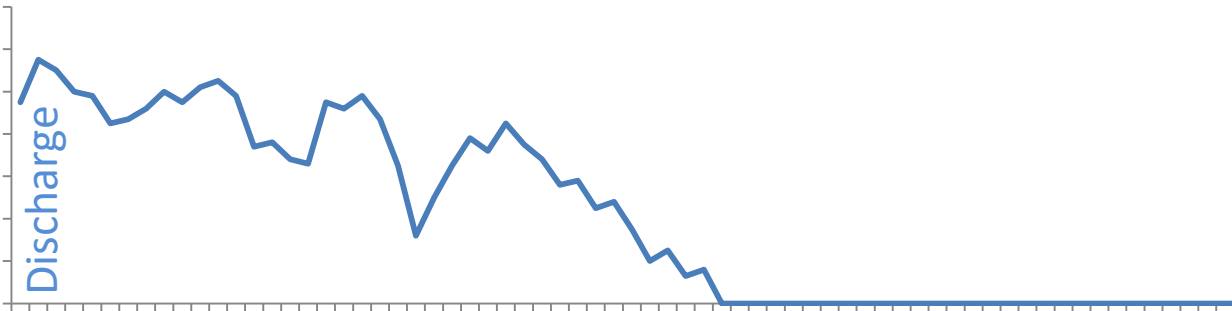
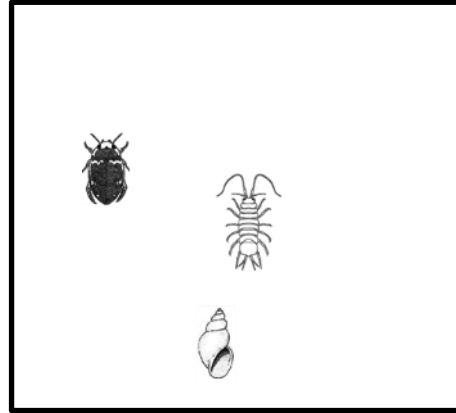
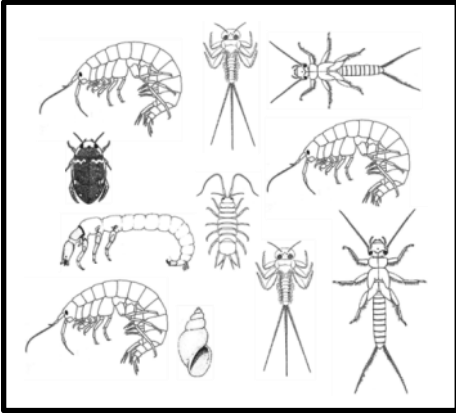
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<sup>b</sup> Department of Botany and Zoology, Faculty of Science, Masaryk University, Kotlářská 2, 611 37 Brno, Czech Republic  
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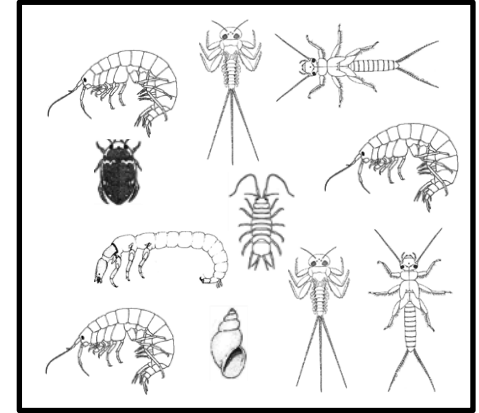
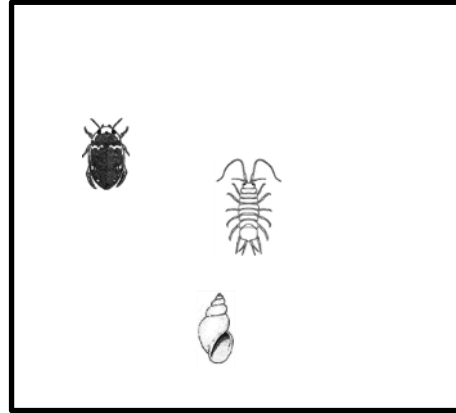
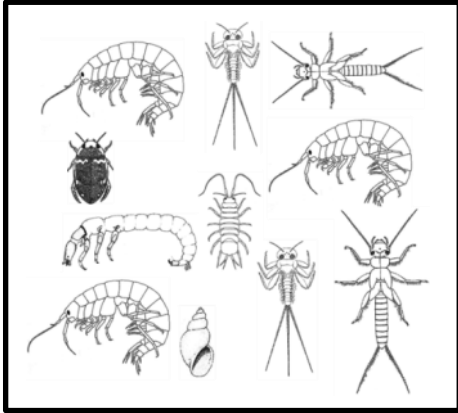
# Theoretical background



# Theoretical background

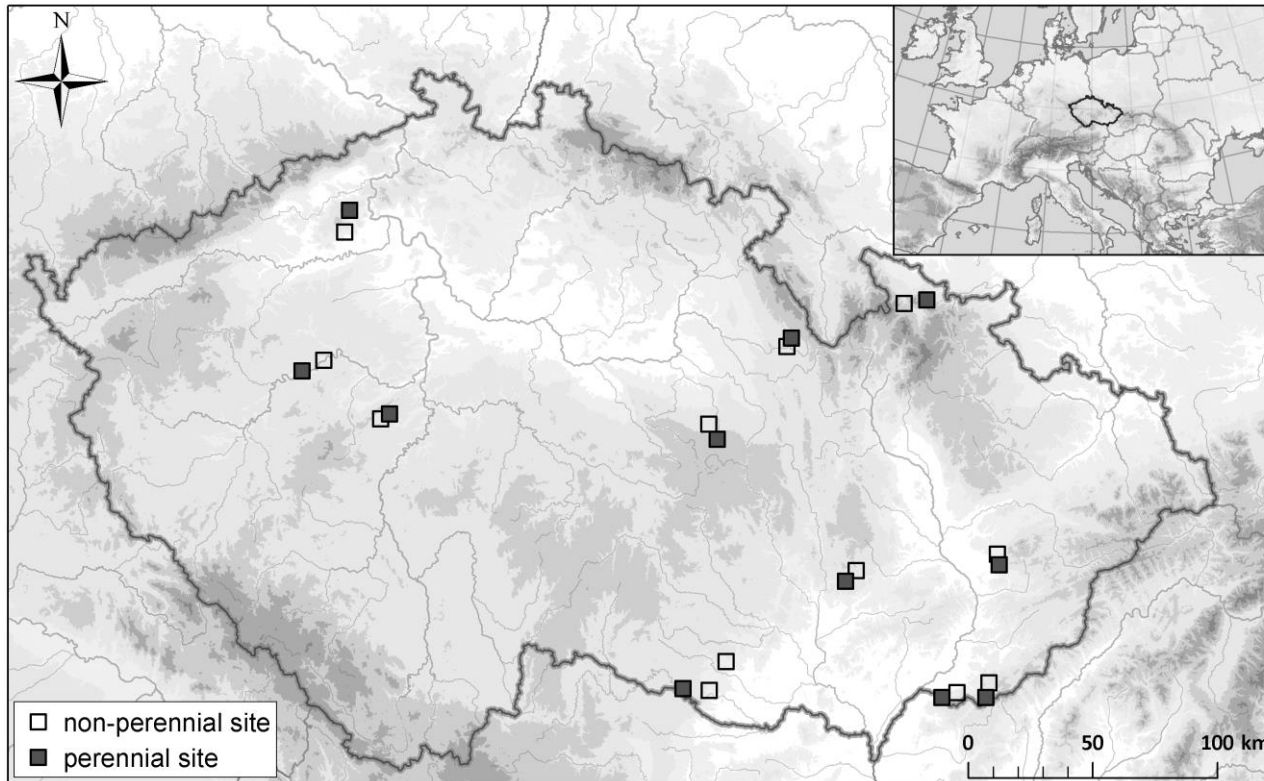


# Theoretical background



# Sites and material

- human unimpacted pairs of intermittent and perennial streams
- 3 min. multihabitat sampling in autumn/spring
- species/genus identification



# Three intermittence categories

## — perennial

- samples taken after > 365 days of continual flow



## — near-perennial

- short flow abruption < 1 week
- dry stretches only hundred meters
- isolated pools occasionally present
- irregular drying



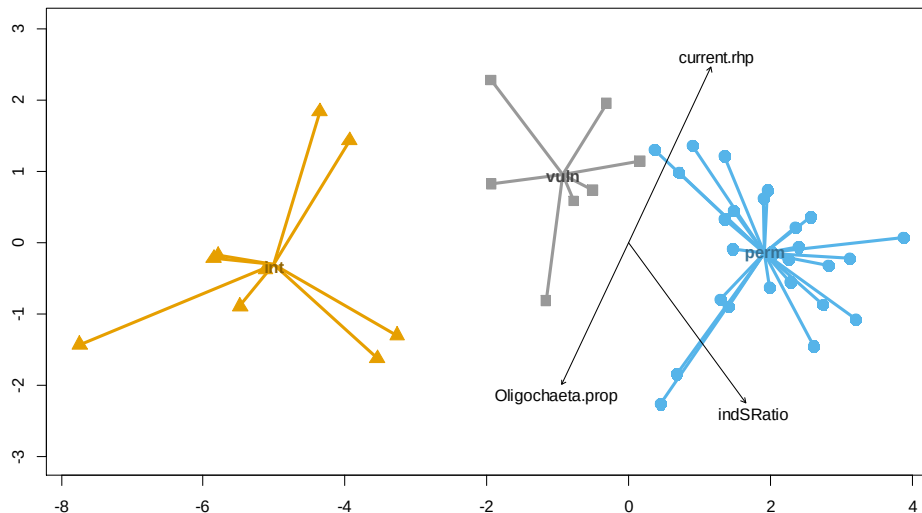
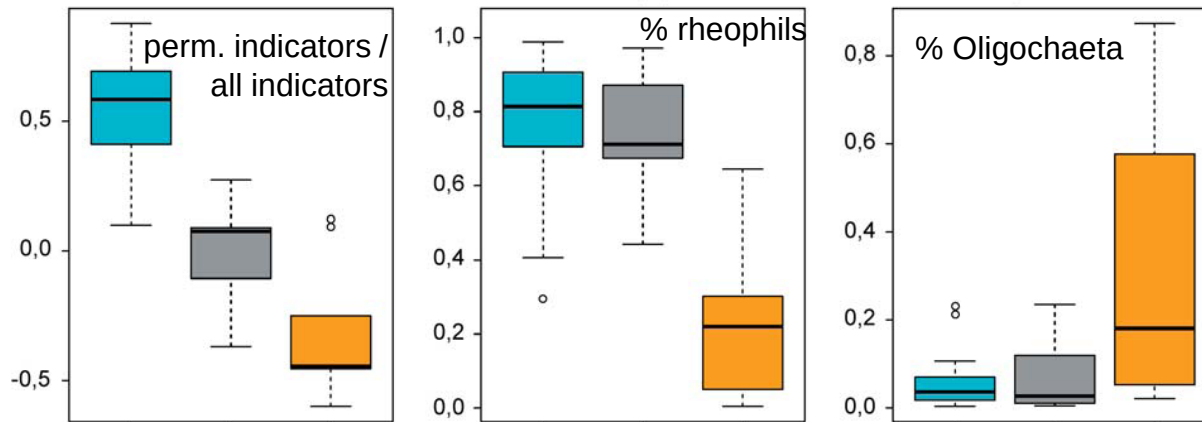
## — intermittent

- long flow abruption > 1 week
- dry stretches > 1 km



# Linear discriminant analyses

- was used to find the metrics with best discrimination power



Final formula for calculating discrimination score was successfully derived:

$$\text{Biodrought index} = (4.09 * \text{Indicator\_taxa}) + (-1.76 * \text{High\_body\_flexibility}) + (0.47 * \text{Total\_abundance}) - 4.01$$

# Biodrought index: final metric used for calculation

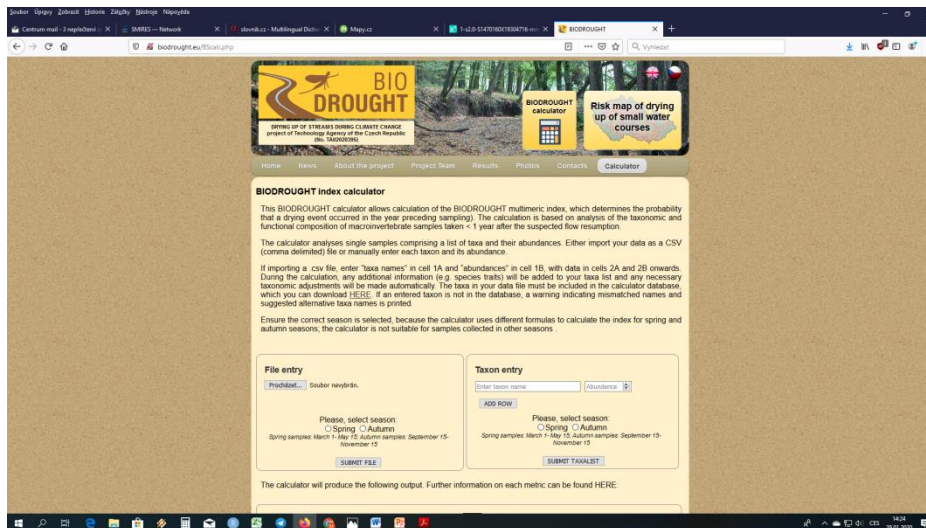
- Proportion of indicator taxa (intermittent vs. perennial) - IT
- Proportion of taxa with high body flexibility - HBF
- Preference for organic substrates - POS
- Total abundace - TA

Autumn sample:

$$\text{Biodrought index} = (4.23 * IT) + (-1.69 * HBF) + (-2.94 * POS) - 0.26$$

Spring sample:

$$\text{Biodrought index} = (4.09 * IT) + (-1.76 * HBF) + (0.47 * TA) - 4.01$$



The screenshot shows the Biodrought calculator web interface. It features a header with the Biodrought logo and navigation links. The main content area includes a 'Biodrought index calculator' section with instructions on how to use the tool, including file entry and taxon entry options. The interface is designed to be user-friendly, with clear instructions and a structured layout for data entry.

| Taxon                            | Reliability | IV Spring/Autumn | Significance of IV Spring/Autumn | Frequency Spring/Autumn | References                                                                                                     |
|----------------------------------|-------------|------------------|----------------------------------|-------------------------|----------------------------------------------------------------------------------------------------------------|
| <b>Platyhelminthes</b>           |             |                  |                                  |                         |                                                                                                                |
| <i>Mesostoma lingua</i>          | 1           |                  |                                  |                         | Young, 2001                                                                                                    |
| <b>Nematomorpha</b>              | 1           | -/0.06           | -/n.s.                           | 0/0.06                  | Clifford, 1966                                                                                                 |
| <i>Gordiidae</i>                 | 1           |                  |                                  |                         | Clifford, 1966                                                                                                 |
| <i>Gordius</i> sp.               | 1           |                  |                                  |                         | Clifford, 1966                                                                                                 |
| <b>Mollusca</b>                  |             |                  |                                  |                         |                                                                                                                |
| <i>Anisus leucostoma</i>         | 2           |                  |                                  |                         | Piechocki, 1979; Jurkiewicz-Karnkowska, 2011; Costil et al., 2001                                              |
| <i>Anisus septemgyratus</i>      | 2           |                  |                                  |                         | Jurkiewicz-Karnkowska, 2008; Jurkiewicz-Karnkowska, 2011                                                       |
| <i>Anisus</i> sp.                | 1           |                  |                                  |                         | Costil et al., 2001                                                                                            |
| <i>Anisus spirorbis</i>          | 1           |                  |                                  |                         | Jurkiewicz-Karnkowska, 2008                                                                                    |
| <i>Aplexa hypnorum</i>           | 2           |                  |                                  |                         | Jurkiewicz-Karnkowska, 2008; Costil et al., 2001                                                               |
| <i>Bathymphalus cantartus</i>    | 1           |                  |                                  |                         | Jurkiewicz-Karnkowska, 2011                                                                                    |
| <i>Bithynia leachii</i>          | 2           |                  |                                  |                         | Jurkiewicz-Karnkowska, 2008; Piechocki, 1979                                                                   |
| <i>Bythinella austriaca</i> agg. | 2           | 0.44/0.46        | */n.s.                           | 0.45/0.56               | Meyer and Meyer, 2000; Acuña et al., 2005; Stubbington et al., 2009; Costil et al., 2001; Falkner et al., 2001 |
| <i>Galba truncatula</i>          | 2           | 0.24/0.18        | n.s./n.s.                        | 0.27/0.19               | Jurkiewicz-Karnkowska, 2008                                                                                    |
| <i>Gyraulus riparius</i>         | 1           |                  |                                  |                         | Piechocki and Dyduch-Falniowska, 1993                                                                          |
| <i>Musculium lacustre</i>        | 1           |                  |                                  |                         | Piechocki and Dyduch-Falniowska, 1993                                                                          |
| <i>Pisidium casertanum</i>       | 1           |                  |                                  |                         | Piechocki and Dyduch-Falniowska, 1993                                                                          |
| <i>Pisidium obtusale</i>         | 1           |                  |                                  |                         | Piechocki and Dyduch-Falniowska, 1993                                                                          |
| <i>Pisidium personatum</i>       | 1           |                  |                                  |                         | Jurkiewicz-Karnkowska, 2011                                                                                    |
| <i>Planorbis carneus</i>         | 1           |                  |                                  |                         | Jurkiewicz-Karnkowska, 2011; Falkner et al., 2001                                                              |
| <i>Planorbis planorbis</i>       | 2           |                  |                                  |                         | Costil et al., 2001                                                                                            |
| <i>Potamopyrgus antipodarum</i>  | 1           | 0.09/-           | n.s./-                           | 0.09/0.06               | Costil et al., 2001; Extence, 1981                                                                             |
| <i>Radix peregra</i>             | 2           | -/0.06           | -/n.s.                           | 0/0.06                  | Jurkiewicz-Karnkowska, 2011; Falkner et al., 2001                                                              |
| <i>Segmentina nitida</i>         | 2           |                  |                                  |                         | Falkner et al., 2001                                                                                           |
| <i>Stagnicola fuscus</i>         | 1           |                  |                                  |                         | Falkner et al., 2001                                                                                           |
| <i>Stagnicola occulta</i>        | 1           |                  |                                  |                         | Jurkiewicz-Karnkowska, 2011                                                                                    |
| <i>Stagnicola palustris</i>      | 1           |                  |                                  |                         | Jurkiewicz-Karnkowska, 2008; Jurkiewicz-Karnkowska, 2011; Falkner et al., 2001                                 |
| <i>Valvata cristata</i>          | 2           |                  |                                  |                         | Jurkiewicz-Karnkowska, 2008; Jurkiewicz-Karnkowska, 2011; Falkner et al., 2001                                 |
| <i>Valvata macrostoma</i>        | 2           |                  |                                  |                         | Fenoglio et al., 2007; Hynes, 1958; Dumnicka and Koszałka, 2005                                                |
| <b>Oligochaeta</b>               |             |                  |                                  |                         |                                                                                                                |
| <i>Eiseniella tetraedra</i>      | 2           | 0.54/0.64        | n.s./*                           | 0.82/0.75               | Fenoglio et al., 2007; Hynes, 1958; Dumnicka and Koszałka, 2005                                                |
| <i>Enchytraeus</i> sp.           | 2           | 0.24/0.25        | n.s./*                           | 0.27/0.25               |                                                                                                                |
| <i>Epirodillus pygmaeus</i>      | 1           | 0.18/0.2         | n.s./n.s.                        | 0.18/0.25               |                                                                                                                |
| <i>Haplotaxis gordioides</i>     | 2           | 0.29/0.32        | n.s./*                           | 0.36/0.38               |                                                                                                                |
| <i>Marionina</i> sp.             | 2           | 0.61/0.12        | */n.s.                           | 0.73/0.13               |                                                                                                                |
| <i>Nois elinguis</i>             | 2           | 0.18/0.3         | n.s./**                          | 0.18/0.31               |                                                                                                                |
| <i>Stylodrilus heringianus</i>   | 2           | 0.56/0.52        | n.s./n.s.                        | 0.64/0.63               | Dumnicka and Koszałka, 2005; Nijboer et al., 2004                                                              |

# Practical use of BIODROUGHT index

Lack of information about stream flow regime



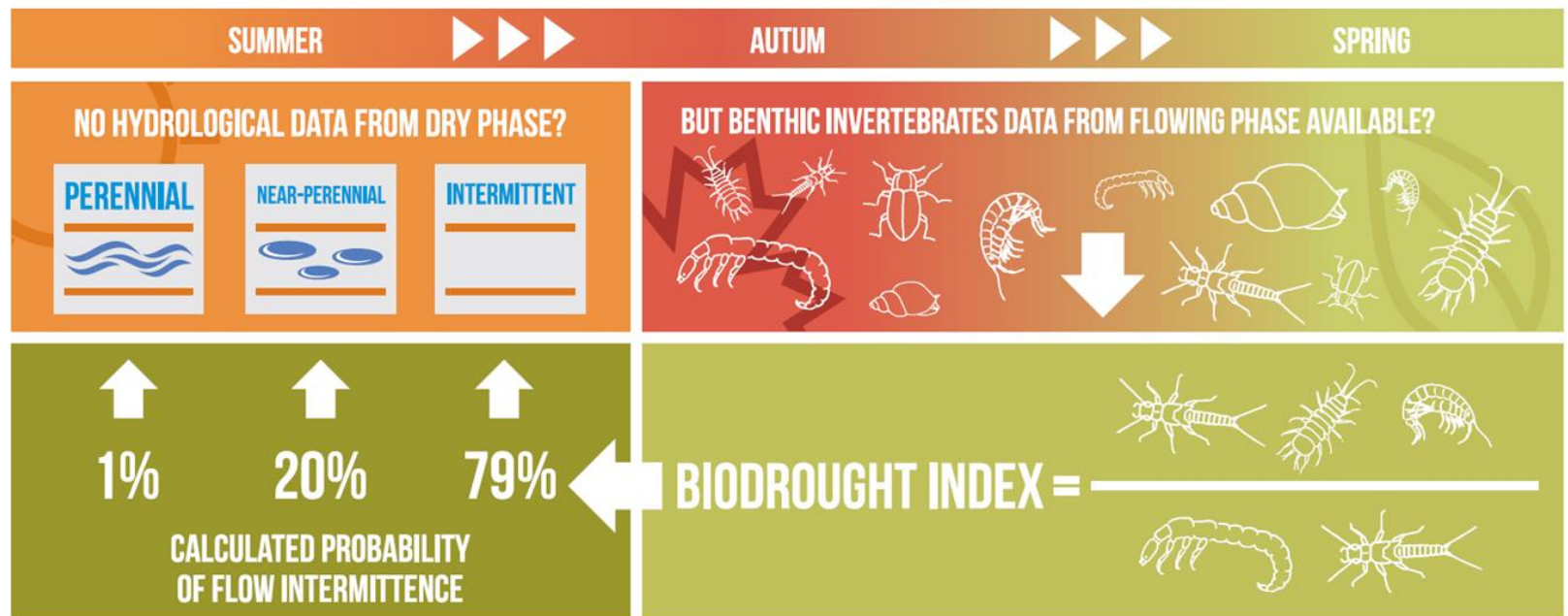
Analyses of benthic invertebrate sample



Assessment of presence of dry epizode with BIODROUGHT method

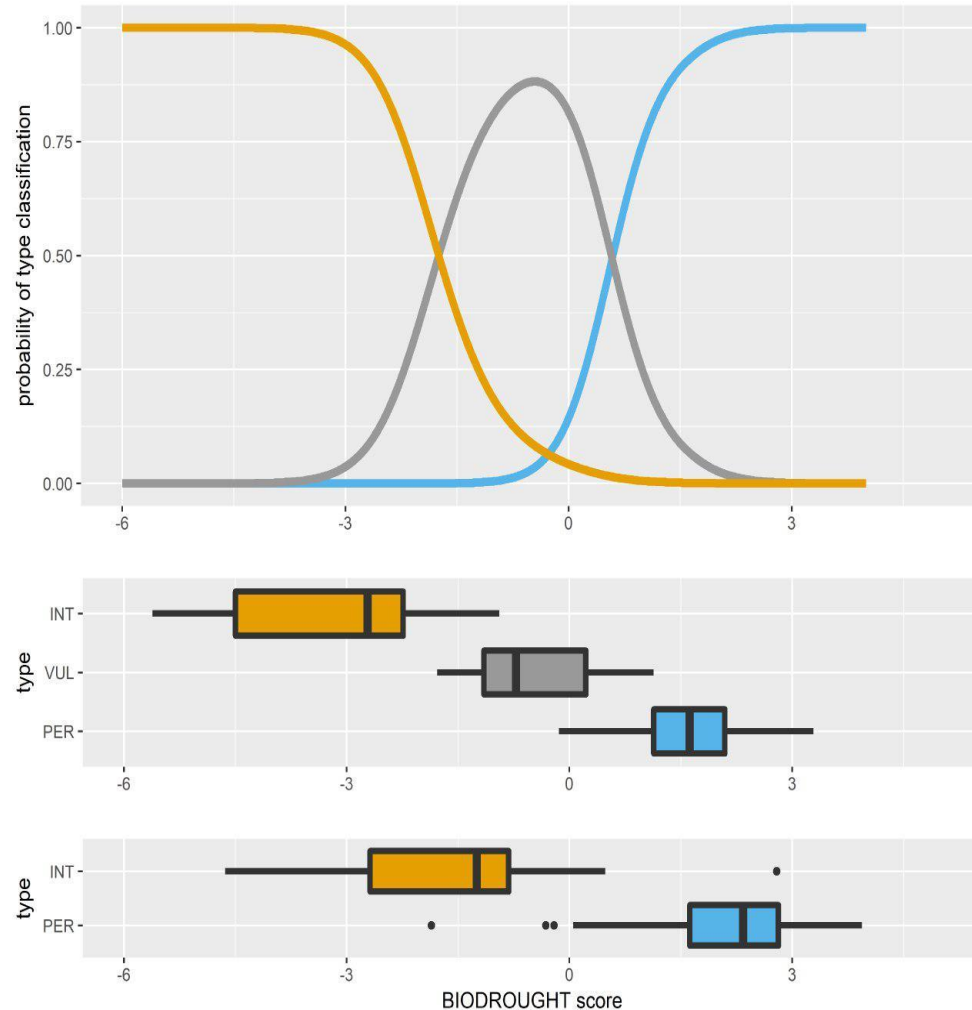
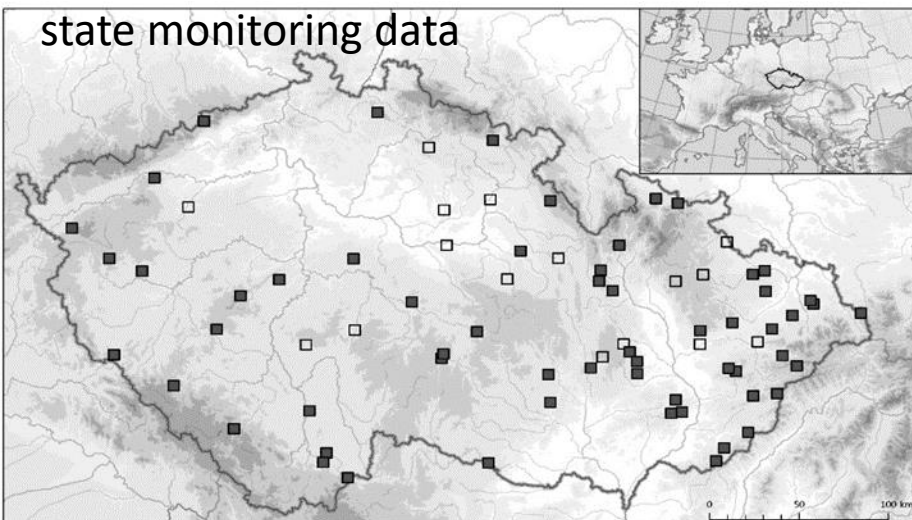


Action, e.g. restoration measures; new assessment of ecological status; revision of water abstraction licence



# Biodrought index

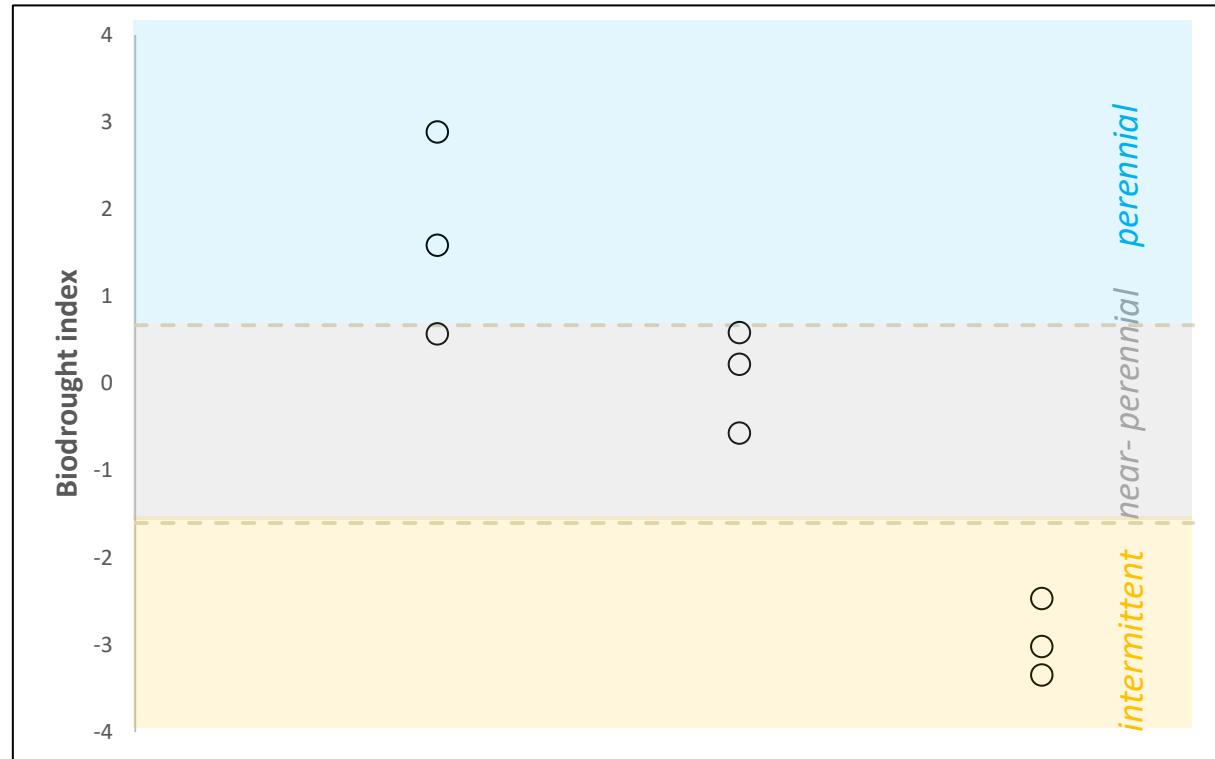
- enables to calculate probability that stream was dry in last year
- in independent dataset was correctly recognised >90% sites



# Biodrought index testing at wider geographical scale



**Austria - Styria**



Dataset from: O. Zweidick, W. Graf

perennial    near-perennial    intermittent



observed instream conditions

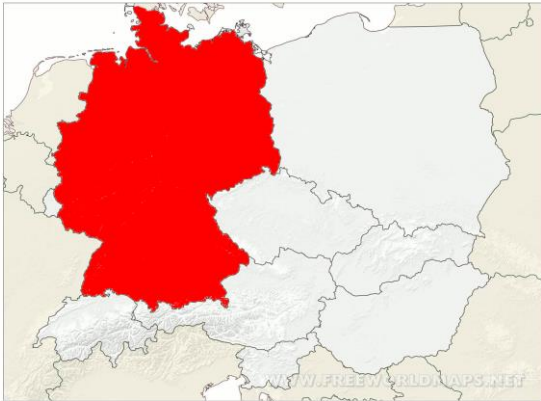


Universität für Bodenkultur Wien  
University of Natural Resources  
and Life Sciences, Vienna

Biodrought index correctly recognized: 89%

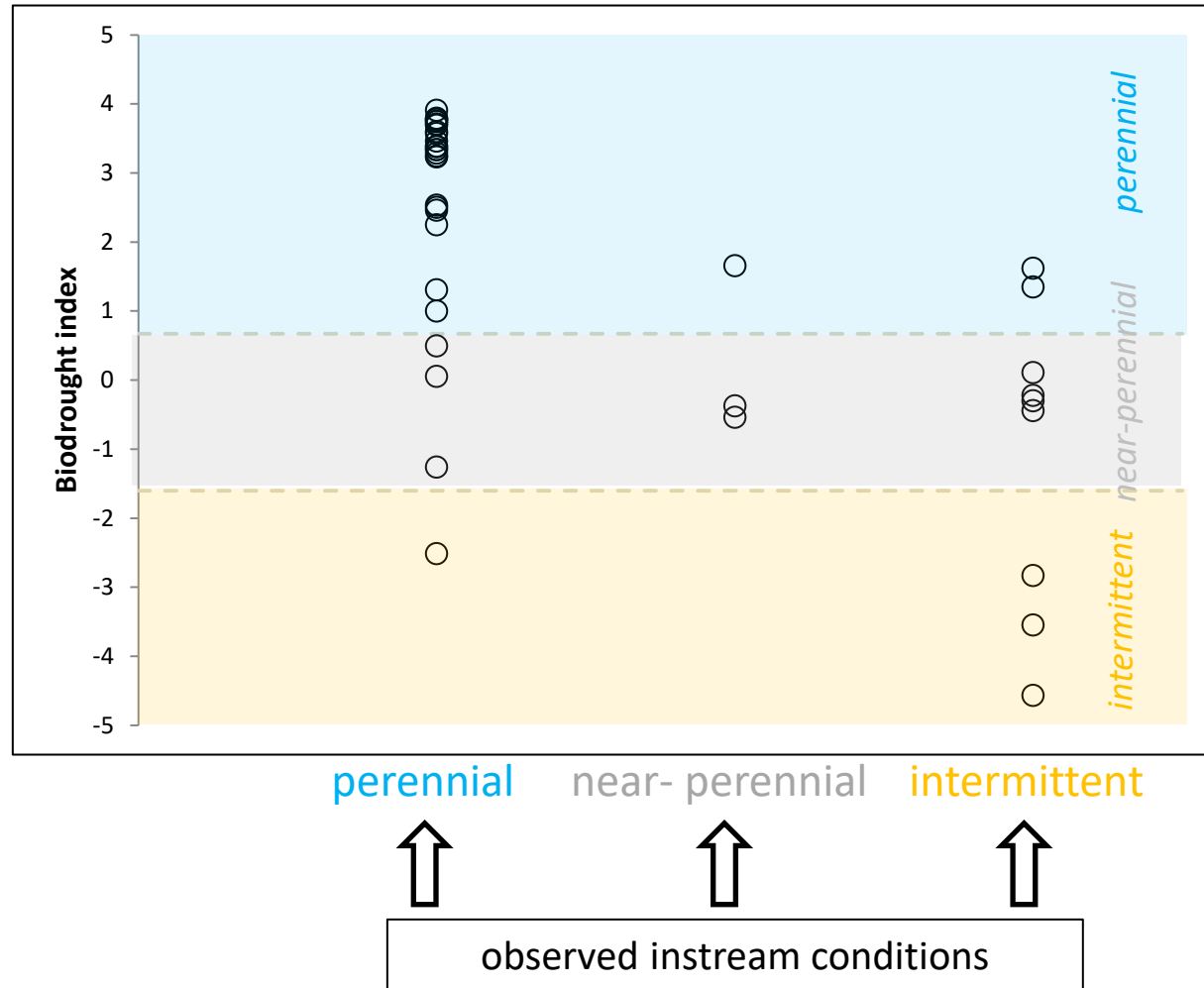
Perennial sample misinterpreted as intermittent and vice versa: 0%

# Biodrought index testing at wider geographical scale



**Germany** – karst streams in East Westphalia

Dataset from: E. I. Meyer



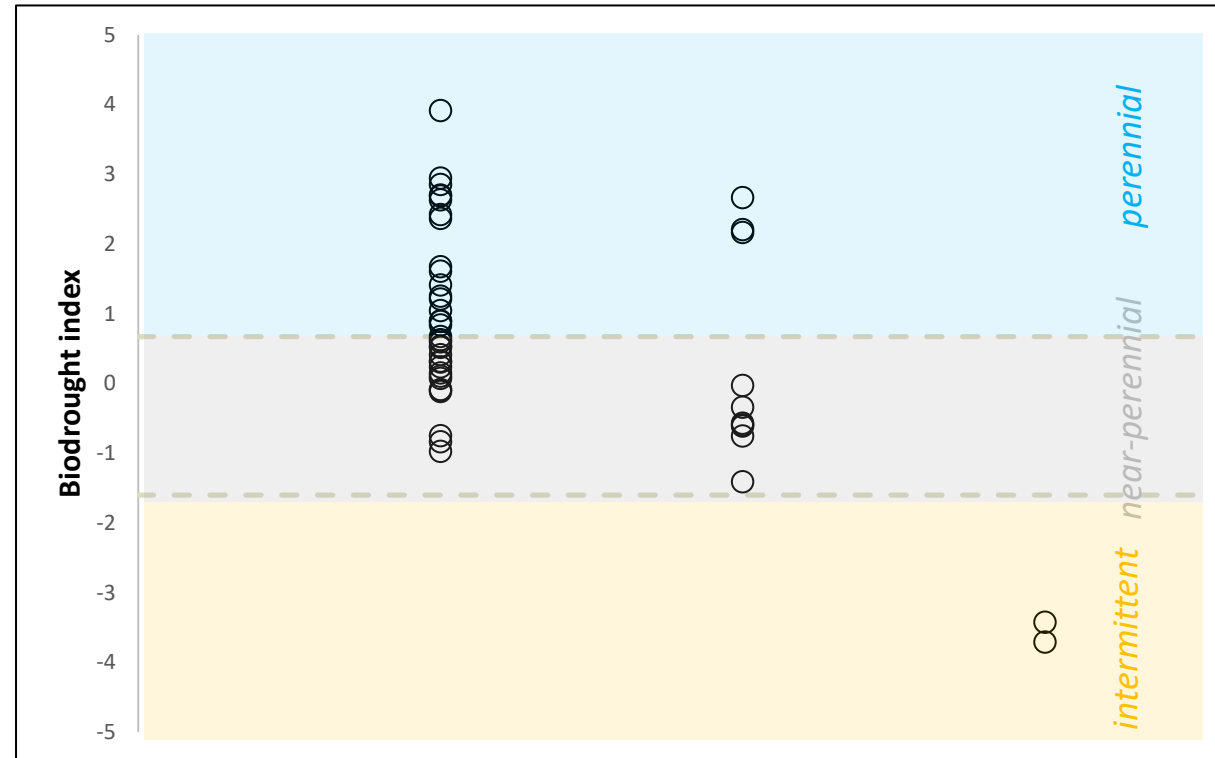
Biodrought index correctly recognized: 73%

Perennial sample misinterpreted as intermittent and vice versa: 6%

# Biodrought index testing at wider geographical scale



Hungary - Mecsek Mts.



Dataset from: Z. Csabai



perennial near-perennial intermittent



observed instream conditions

Biodrought index correctly recognized: 57%

Perennial sample misinterpreted as intermittent and vice versa: 0%



Thank you for your  
attention

[www.biodrought.eu](http://www.biodrought.eu)